

Topic 1B: Hallucination, Calibration, and Missing Mass

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Acknowledgements: These notes draw on Kalai and Vempala [1], Good [2], Kleinberg and Mullainathan [5], Kalavasis, Mehrotra, and Velegkas [6], Miao and Kearns [7], Kalai’s Simons Institute lecture [8], and the subsequent work of Kalai, Nachum, Vempala, and Zhang [9].

AI Disclosure: These notes were drafted and revised with the assistance of AI tools. I have reviewed the content and take responsibility for it. If you find an error, please let me know.

In the previous lecture, success meant producing one new valid output. A learner could find a safe part of the language and remain there. A model that tries to reproduce a source distribution faces a stronger requirement. It must learn not only which outputs are valid, but also how often the source produces them.

Imagine a collection of university documents that mention courses. Half of the mentions refer to *Introduction to Programming*. The remaining mentions are spread across many specialized courses, most of which appear only once in the sample. A generative model that reproduces this source cannot simply repeat the common course. It must spread about half of its probability across the long tail of specialized courses. Copying titles from the sample is safe, but it misses real courses that were not observed. Placing probability on plausible unseen titles better represents the source, but some of those titles may not name real courses.

Kalai and Vempala [1] turn this conflict into a statistical question. Suppose every training document is correct, the documents are independent draws from a fixed distribution, and computation is unlimited. *Can learning the source distribution itself force a model to hallucinate?* For facts whose unseen identities cannot be inferred from structure, their answer is yes. The resulting hallucination rate is controlled by a statistic that we can estimate from the training data.

1 A Statistical Model of Factual Generation

We represent each document by the single factual claim it communicates. Different wordings of the same claim correspond to the same *factoid*.

As in the previous lecture, $\mathcal{U}$ denotes the universe of possible outputs, and the target language $L^* \subseteq \mathcal{U}$ contains the valid ones, here the true factoids. Unlike the previous lecture, $\mathcal{U}$ is finite, although possibly enormous, because we will need to count how many candidates are true and false. Let

$$\mathcal{H} = \mathcal{U} \setminus L^*$$

be the set of plausible but false factoids. We say that a generator *hallucinates* when it produces an output in $\mathcal{H}$.

The data. Documents need not mention every true factoid equally often. A source distribution p describes these frequencies. It assigns positive probability to every true factoid and zero probability to every false one:

$$p(x) > 0 \text{ for } x \in L^*, \quad p(x) = 0 \text{ for } x \in \mathcal{H}.$$

Together, L^* and p specify which factoids are true and how often documents mention them. The learner does not know either one. It sees n documents sampled independently from p . If X_i is the factoid in document i , write $\mathbf{X} = (X_1, \dots, X_n)$ for the full sample and let

$$S = \{X_1, \dots, X_n\}$$

be the set of distinct factoids in the training data. In the previous lecture, we used S_t for the examples observed by round t . Here all n documents arrive as one sample, so we write S .

Since p places probability only on truths, every factoid in S is true, and hence $S \subseteq L^*$. We write $\bar{S} = \mathcal{U} \setminus S$ for the factoids that did not appear. This set contains both true factoids that the sample missed and false factoids in $\mathcal{H}$. Figure 1 shows the distinction that the learner cannot see.

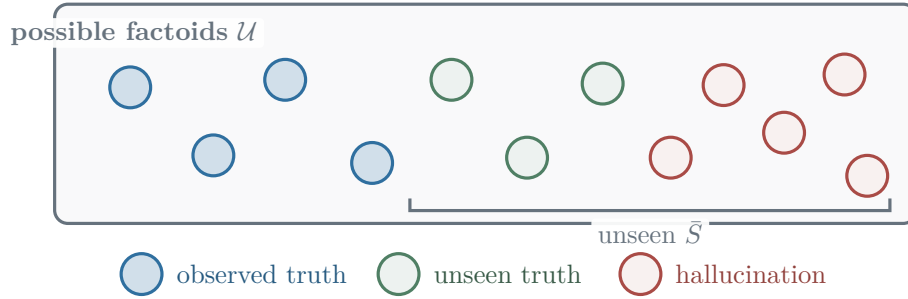


Figure 1: The learner sees the blue factoids. It does not know which of the remaining candidates are unseen truths and which are hallucinations.

The generator. After seeing the n documents, the learner constructs a distribution q over $\mathcal{U}$, where $q(x)$ is the probability that it generates factoid x .

2 A Simplified Hallucination Bound

For any set $A \subseteq \mathcal{U}$, write the total probabilities that the source and the generator assign to A as

$$p(A) = \sum_{x \in A} p(x), \quad q(A) = \sum_{x \in A} q(x).$$

In particular, $q(\mathcal{H})$ is the probability that the generator hallucinates. As Figure 1 shows, every unseen factoid $\in \bar{S}$ is either a truth $\in L^* \setminus S$ that the sample missed or a false factoid $\in \mathcal{H}$. Therefore

$$\underbrace{q(\mathcal{H})}_{\text{probability of a hallucination}} = \underbrace{q(\bar{S})}_{\text{probability of an unseen output}} - \underbrace{q(L^* \setminus S)}_{\text{probability of a new valid output}}. \quad (1)$$

A generator can make $q(\bar{S})$ zero by producing only factoids in the sample. It would then avoid hallucination, but it would also ignore every true factoid that the sample missed. If the generator is to represent those missing truths, it must place some probability on $\bar{S}$. Avoiding hallucination then requires that this probability find $L^* \setminus S$, rather than the false factoids in $\mathcal{H}$.

Define the missing mass of $\mathbf{X} = (X_1, \dots, X_n)$ by $M_p(\mathbf{X}) := p(\mathcal{U} \setminus \{X_1, \dots, X_n\})$. Since the source places no probability on false factoids,

$$\underbrace{M_p(\mathbf{X})}_{\text{missing mass}} = p(\bar{S}) = p(L^* \setminus S).$$

We first prove the bound under two simplifying assumptions.

1. **Breadth.** The generator produces an unseen factoid with the same probability as a fresh source document does:

$$q(\bar{S}) = p(\bar{S}) = M_p(\mathbf{X}).$$

This fixes only the total probability outside the sample. It does not require $q = p$, nor does it tell the generator which unseen factoids are true.

2. **Symmetric uncertainty.** Before the documents are drawn, we choose a fixed number of true factoids uniformly from $\mathcal{U}$, then let the source p be uniform over them. The names of unseen factoids therefore carry no clues about which ones are true.

Now condition on the observed sample. Every possible truth set of the chosen size that contains S makes this sample equally likely. The unseen truths are therefore uniformly distributed among the candidates in $\bar{S}$. Exactly $|L^* \setminus S|$ of the $|\bar{S}|$ unseen candidates are true, so this common probability must be

$$\mathbb{P}(x \in L^* \mid \mathbf{X}) = \frac{|L^* \setminus S|}{|\bar{S}|} \quad \text{for every } x \in \bar{S}.$$

Now fix the sample and the resulting generator q . The only remaining randomness is which source p was chosen. A candidate x contributes $q(x)$ to the probability of a new valid output if it is true, and zero otherwise. Adding these expected contributions gives

$$\mathbb{E}[q(L^* \setminus S) \mid \mathbf{X}] = \sum_{x \in \bar{S}} q(x) \mathbb{P}(x \in L^* \mid \mathbf{X}) = \frac{|L^* \setminus S|}{|\bar{S}|} q(\bar{S}).$$

Redistributing probability among the unseen candidates does not change this expectation because all of them have the same posterior probability of being true. Substituting the calculation and $q(\bar{S}) = M_p(\mathbf{X})$ into Equation (1) gives

$$\begin{aligned} \mathbb{E}[q(\mathcal{H}) \mid \mathbf{X}] &= q(\bar{S}) - \mathbb{E}[q(L^* \setminus S) \mid \mathbf{X}] \\ &= \left(1 - \frac{|L^* \setminus S|}{|\bar{S}|}\right) M_p(\mathbf{X}). \end{aligned}$$

The fraction $|L^* \setminus S|/|\bar{S}|$ is the proportion of unseen candidates that are true. This fraction is small when there are many plausible unseen factoids but few of them are true.

Let's return to the course example. Let the set of plausible course titles be $\mathcal{U} = \{A, B, C, D\}$. Choose the two real courses, L^* , uniformly from the six pairs in $\mathcal{U}$, and let the source p assign probability $1/2$ to each. Thus $p(x)$ is the probability that a fresh document mentions course x .

Suppose the sample contains only A . Then $S = \{A\}$ and $\bar{S} = \{B, C, D\}$, and each of B, C, D is equally likely to be the other real course. Hence $|L^* \setminus S|/|\bar{S}| = 1/3$.

After seeing the sample, the learner produces q , where $q(x)$ is the probability that the model outputs course x . The breadth condition requires

$$q(B) + q(C) + q(D) = q(\bar{S}) = p(\bar{S}) = \frac{1}{2}.$$

Thus the expected probability of a new valid course is $(1/3)(1/2) = 1/6$, and the expected hallucination probability is $1/2 - 1/6 = 1/3$.

2.1 How the Full Result Relaxes the Assumptions

The calculation above assumed that the generator and the source place exactly the same total probability on the set of unseen factoids. Kalai and Vempala obtain this kind of breadth from a weaker requirement called *semantic calibration* [1]. To define it, group together all factoids to which the generator assigns the same probability. For each probability value z , let

$$B_z = \{x \in \mathcal{U} : q(x) = z\}.$$

The generator assigns total probability $q(B_z) = z|B_z|$ to this group. It is semantically calibrated if the source assigns the same total probability:

$$p(B_z) = q(B_z) \quad \text{for every nonempty group } B_z.$$

This condition does not require $p(x) = q(x)$ for each factoid. It only compares the total probability of factoids that the generator treats alike.

Why is semantic calibration useful here? A generator could avoid hallucination by producing only observed truths, but it would then ignore the source’s missing mass. Calibration prevents this loss of breadth without requiring the generator to recover the source probability of every individual factoid. In the simple case where no probability group contains both observed and unseen factoids, summing the calibration equalities over the unseen groups gives $q(\bar{S}) = p(\bar{S})$, exactly the breadth equality used above.

The full theorem allows approximate calibration and probability groups that contain both observed and unseen factoids. Any disagreement between the source and the generator weakens the hallucination bound by a corresponding calibration term. Thus a generator can reduce hallucination by concentrating more heavily on observed truths, but doing so makes it less calibrated to the source.

The paper replaces perfect symmetry with a regularity condition. Regularity says that after we observe the sample, no particular unseen factoid is much more likely than the others to be true. This condition plays the same role as symmetry in the calculation above. Separately, the result assumes sparsity: only a small fraction of the plausible factoids are true. Together, these assumptions limit how much of the generator’s probability on unseen outputs can reliably reach a truth. The source probabilities themselves may be unequal. For example, we can assign a fixed long-tailed profile at random to the true factoid names.

3 Estimating Missing Mass with Good–Turing

The hallucination bound above is proportional to the missing mass $M_p(\mathbf{X})$. When $M_p(\mathbf{X})$ is large, breadth requires the generator to place substantial probability on unseen factoids; when it is small, this pressure is weak. We therefore need to estimate $M_p(\mathbf{X})$ from the sample. At first this seems impossible: the quantity depends precisely on factoids that the sample does not contain.

Good–Turing estimation uses factoids that appear exactly once as evidence of unseen source probability. If course mentions repeatedly name the same few courses, the missing mass is likely small; if many course titles appear only once, it is likely larger. Good introduced this idea for species and vocabulary, while crediting Turing for much of its development [2].

Definition 3.1 (Monofact rate). Let $N_1(\mathbf{X})$ be the number of factoids that appear exactly once in the sample. The *monofact rate* is the fraction of document positions occupied by these singleton factoids:

$$\widehat{\text{MF}}(\mathbf{X}) = \frac{N_1(\mathbf{X})}{n}.$$

To see why, choose an index J uniformly from $\{1, \dots, n\}$, independently of $\mathbf{X}$, and set aside document J . Let $\mathbf{X}_{-J} = (X_i)_{i \neq J}$ and

$$S_{-J} = \{X_i : i \neq J\}$$

be the factoids in the remaining $n - 1$ documents. The held-out factoid X_J is absent from S_{-J} exactly when it appeared once in the full sample. Conditional on the full sample, there are exactly $N_1(\mathbf{X})$ choices of J for which this happens. Therefore

$$\mathbb{P}(X_J \notin S_{-J} \mid \mathbf{X}) = \frac{N_1(\mathbf{X})}{n} = \widehat{\text{MF}}(\mathbf{X}).$$

Condition on $J = j$. We can generate the same experiment by first drawing $(X_i)_{i \neq j}$ and then independently drawing $X_j \sim p$. Therefore, conditioning on J and the remaining documents gives

$$\mathbb{P}(X_J \notin S_{-J} \mid J, \mathbf{X}_{-J}) = M_p(\mathbf{X}_{-J}).$$

Taking expectations of the two conditional probabilities gives

$$\mathbb{E}[\widehat{\text{MF}}(\mathbf{X})] = \mathbb{E}[M_p(\mathbf{X}_{-J})].$$

Thus the monofact rate is an unbiased estimate of the missing mass for a sample that is one document shorter. To compare it with the missing mass of the full sample, note that a factoid x is absent from m independent draws with probability $(1 - p(x))^m$, and if it is absent, it contributes $p(x)$ to the missing mass. Hence

$$\begin{aligned} \mathbb{E}[M_p(\mathbf{X})] &= \sum_{x \in \mathcal{U}} p(x)(1 - p(x))^n, \\ \mathbb{E}[M_p(\mathbf{X}_{-J})] &= \sum_{x \in \mathcal{U}} p(x)(1 - p(x))^{n-1}, \\ 0 \leq \mathbb{E}[M_p(\mathbf{X}_{-J})] - \mathbb{E}[M_p(\mathbf{X})] &= \sum_{x \in \mathcal{U}} p(x)((1 - p(x))^{n-1} - (1 - p(x))^n) \\ &= \sum_{x \in \mathcal{U}} p(x)^2(1 - p(x))^{n-1} \\ &\leq \frac{1}{n} \sum_{x \in \mathcal{U}} p(x) = \frac{1}{n}, \end{aligned}$$

where the inequality uses $t(1 - t)^{n-1} \leq 1/n$. Concentration bounds then show that the realized monofact rate and missing mass are close with high probability. We use only the direction that lower-bounds the missing mass.

Theorem 3.2 (Good–Turing missing-mass estimate [3, 4]). *For every distribution p , every $n \geq 1$, and every $\delta \in (0, 1/3]$, if $\mathbf{X} \sim p^{\otimes n}$, then with probability at least $1 - \delta$ over $\mathbf{X}$,*

$$M_p(\mathbf{X}) \geq \widehat{\text{MF}}(\mathbf{X}) - \sqrt{\frac{6 \ln(2/\delta)}{n}}.$$

The guarantee is distribution-free. It assumes neither a known number of true factoids nor any particular shape for their probabilities. If 1,000 factoid occurrences contain 180 singletons, then $\widehat{\text{MF}}(\mathbf{X}) = 0.18$. We estimate that the next source draw has roughly an 18% chance of being a factoid absent from the sample. It does not say that 18% of all true factoids remain undiscovered.

Returning to the simplified bound, the theorem lets us replace the unknown missing mass by a quantity computed from the sample. With probability at least $1 - \delta$,

$$\mathbb{E}[q(\mathcal{H}) \mid \mathbf{X}] \geq \left(1 - \frac{|L^* \setminus S|}{|\bar{S}|}\right) \left(\widehat{\text{MF}}(\mathbf{X}) - \sqrt{\frac{6 \ln(2/\delta)}{n}}\right).$$

The first factor is the fraction of unseen candidates that are false. The second is a lower estimate of how much source probability lies outside the sample.

4 Breadth, Validity, and the Tradeoff

Kleinberg and Mullainathan’s positive result provides validity without requiring the generator to move beyond a safe subset [5]. Kalavasis, Mehrotra, and Velegkas use *breadth* for the additional requirement that a generator cover the full range of valid outputs [6].

In our statistical setting, the source distribution gives a natural version of breadth. A generator should represent the range of factoids that the source can produce, including factoids that happened not to appear in the sample. Our simplified proof expresses this requirement directly through $q(\bar{S}) = p(\bar{S})$. Semantic calibration is one way to enforce this connection between the two distributions.

The lower bound leaves two responses to this pressure. A generator can retain breadth and learn to distinguish the unseen truths in $L^* \setminus S$ from the hallucinations in $\mathcal{H}$. Alternatively, it can concentrate on familiar truths and give up breadth.

4.1 Learning to Recognize Validity

Kalai, Nachum, Vempala, and Zhang relate the hallucination rate of any generator to a classification problem for validity [9]. Consider the following balanced problem, called *Is-It-Valid*. With probability $1/2$, we draw x from the source distribution p and label it valid. With probability $1/2$, we draw x uniformly from $\mathcal{H}$ and label it invalid. The first half contains truths in their natural source frequencies. The second half tests whether we can distinguish those truths from hallucinations.

A generator q defines a classifier that accepts

$$A_q = \left\{x \in \mathcal{U} : q(x) > \frac{1}{|\mathcal{H}|}\right\}.$$

Let $\bar{A}_q = \mathcal{U} \setminus A_q$ be the rejected set. The classification error is

$$\text{IIVErr}(q) = \underbrace{\frac{1}{2}p(\bar{A}_q)}_{\text{true factoid rejected}} + \underbrace{\frac{1}{2}\frac{|A_q \cap \mathcal{H}|}{|\mathcal{H}|}}_{\text{hallucination accepted}}. \quad (2)$$

Define the *threshold discrepancy*

$$\Delta_q := |p(\bar{A}_q) - q(\bar{A}_q)| = |p(A_q) - q(A_q)|.$$

It measures disagreement between p and q only at this threshold.

Proposition 4.1 (Hallucination and validity testing). *For every source distribution p supported on L^* and every generator q ,*

$$q(\mathcal{H}) \geq 2 \text{IIVErr}(q) - \frac{|L^*|}{|\mathcal{H}|} - \Delta_q. \quad (3)$$

The intuition is simple. Every accepted hallucination receives probability more than $1/|\mathcal{H}|$, so accepting many hallucinations forces $q(\mathcal{H})$ to be large. Every rejected truth receives probability at most $1/|\mathcal{H}|$, so all rejected truths together receive at most $|L^*|/|\mathcal{H}|$ from q . Finally, Δ_q compares the source rejection probability $p(\bar{A}_q)$ with the generator rejection probability $q(\bar{A}_q)$. Combining these observations gives

$$2 \text{IIVErr}(q) \leq q(A_q \cap \mathcal{H}) + q(\bar{A}_q \cap \mathcal{H}) + \frac{|L^*|}{|\mathcal{H}|} + \Delta_q = q(\mathcal{H}) + \frac{|L^*|}{|\mathcal{H}|} + \Delta_q,$$

because $A_q \cap \mathcal{H}$ and $\bar{A}_q \cap \mathcal{H}$ partition $\mathcal{H}$. Rearranging gives the proposition. Thus, when $|L^*|/|\mathcal{H}|$ is small and validity is difficult to recognize, $q(\mathcal{H})$ and Δ_q cannot both be small.

Semantic calibration is stronger than the condition used here. Since A_q is a union of level sets of q , semantic calibration implies $\Delta_q = 0$. In contrast, the breadth equality $p(\bar{S}) = q(\bar{S})$ concerns a different set and does not control Δ_q .

4.2 Trading Calibration for Fewer Hallucinations

Miao and Kearns study one way to move along this tradeoff while keeping the monofact rate fixed [7]. In the experiment shown in Figure 2, they first fine-tune a model normally. Late in training, they begin repeating 5% of the examples ten times. The intervention reveals no new truths, so it only shifts probability toward facts already in the sample.

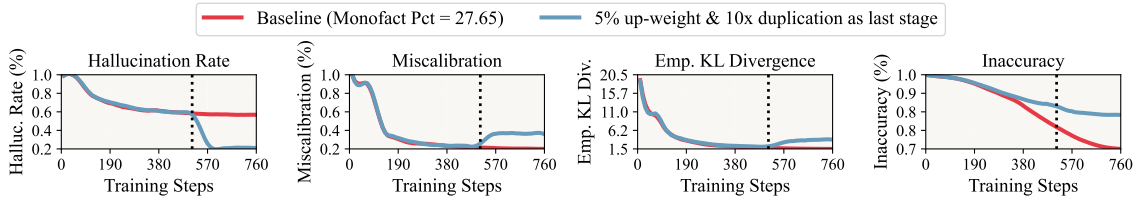


Figure 2: Late-stage upweighting in a fine-tuned T5-Small model. The dashed line marks the start of the intervention. Hallucination falls sharply, while miscalibration and empirical KL divergence rise. Prompted inaccuracy remains near its level before the intervention. Reproduced from Miao and Kearns [7], Figure 6.

Because the intervention provides no information about unseen truths, it reduces hallucination by concentrating the model on familiar truths and moving it farther from the source distribution. Miao and Kearns measure this discrepancy across probability bins, rather than only on A_q . The effect depends on the training stage and model architecture, so selective repetition is evidence for the tradeoff rather than a universal prescription.

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